

CIVL4340

Wind Engineering

Week 7: Wind loading standards (wind climate and ABL definition)

Summary – so far

- Introduction to severe wind events,
 - Wind climate (design wind speeds),
 - Atmospheric boundary layer structure (engineering models),
 - Turbulence,
 - Non-homogeneous ABLs (terrain change, topography).
-
- How can this be codified?

Overview

- AS/NZS1170.2 – revision and examples
- Other standards
 - AS4055 – Housing
 - AS7000 – Transmission lines
 - ISO4354 – Wind actions on structures (International)

Designing for wind

The goal

“The wind load that is required for the design of cladding or a structural frame is one that will produce, to a close approximation, the peak (maximum or minimum) values of the loads or load effects....in a given time period.”

Quasi-steady approach

“[The approach] approximates the peak pressures of the building surface by multiplying a quasi-steady pressure coefficient, with a dynamic pressure based on the peak design gust wind speed”

- Assumes max wind generates max load

Designing for wind

AS/NZS1170.2 (2021): Structural design actions Part 2: Wind actions

“The objective of this Standard is to provide wind actions for use in the design of structures subject to wind action. It provides a detailed procedure for the determination of wind actions on structures, varying from those less sensitive to wind action to those for which dynamic response must be taken into consideration.”

AS4055 (2021): Wind loads for housing

“The objective of this document is to provide designers, builders and manufacturers of building products that are affected by wind loading with a range of wind speed classes that can be used to design and specify such products for use in housing that are within the limitations of this document”

Designing for wind

AS/NZS7000 (2016): Overhead line design – Appendix B

*“This Standard specifies the general requirements that shall be met for the design and construction of **new overhead lines** to ensure that the line is suitable for its intended purpose...”*

ISO4354 (2009): Wind actions on structures

“This International Standard is intended for use by countries without an adequate wind loading standard and as a bridge between existing International Standards.”

“This International Standard describes the actions of wind on structures and specifies methods of calculating characteristic values of wind loads for use in designing buildings, towers, chimneys, bridges and other structures, as well as their components and appendages.....In particular, this International Standard facilitates the conversion between peak and mean wind speed methodologies and covers the three main storm types, synoptic winds, thunderstorms and tropical cyclones (hurricanes and typhoons)”

AS/NZS1170.2 - overview

Determine design wind speed and wind actions on structures (within limits) other than those caused by tornadoes.

Limits:

Buildings < 200 m tall

Unsupported roof span < 100 m

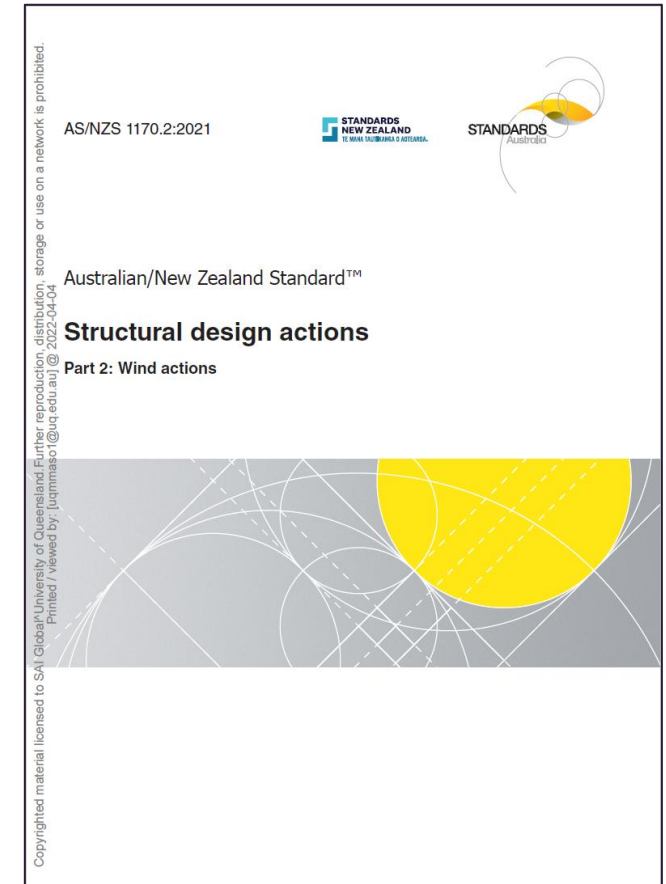
Offshore structures within 30 km from nearest coastline

NOT offshore structures, bridges, windfarm structures, power transmission and distribution structures

Natural frequency < 1 Hz needs dynamic analysis

Outside limits – seek alternate solutions (wind tunnel, CFD) or use of international standards (e.g. wind turbines)

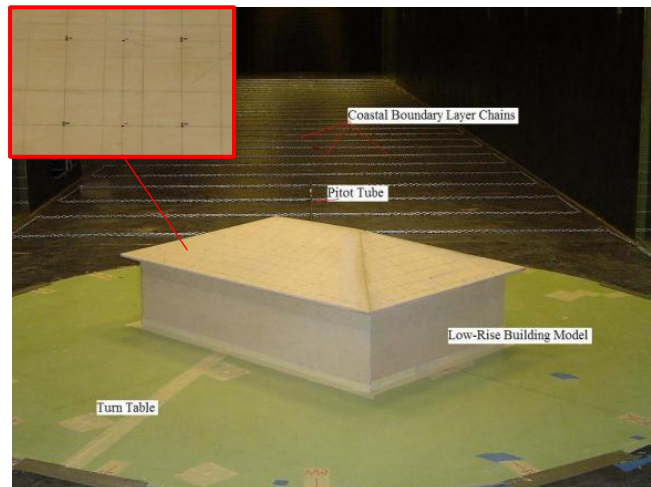
Engineering judgment is required to decide whether the Standard can be applied.



Alternate solutions

Wind tunnel testing

- Exact replication of structural form at reduced scale.
- Need to replicate building surrounds so that the wind speed and turbulence profiles also scales.
- Scaling of, Length, Velocity, Time.
- Generates explicit (generally) pressure or dynamic loads on structures that can be implemented in design calculations.



Goliber 2009

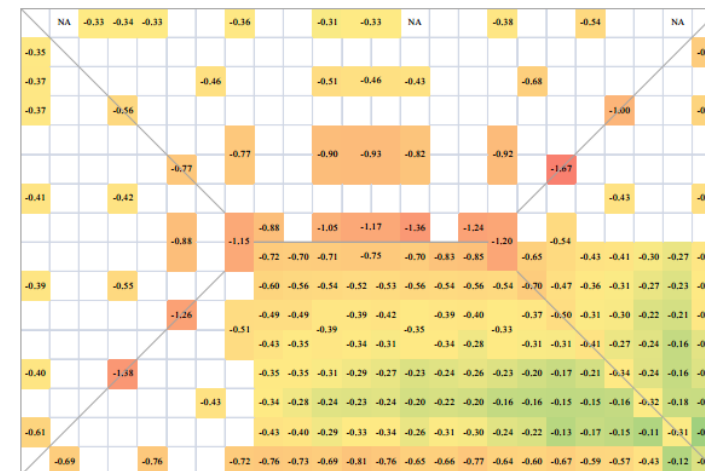
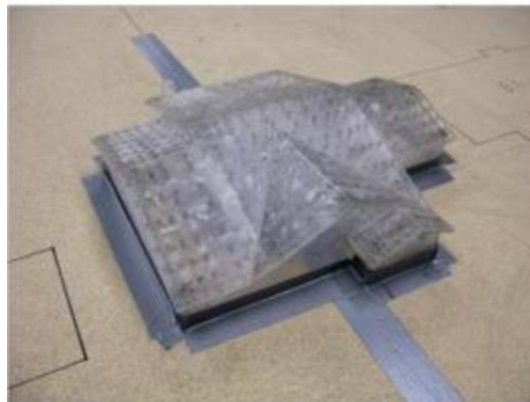


Figure 6.21. Pressure coefficients on the roof of Model 4 with a wind attack orientation of 30°

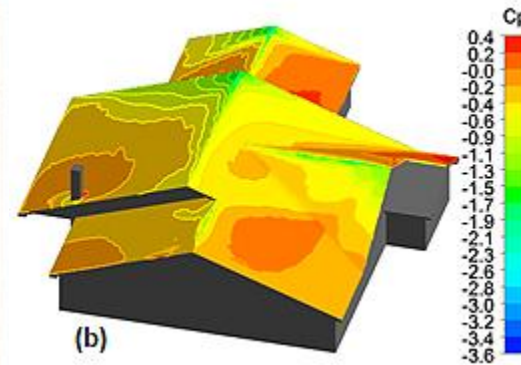
Alternate solutions

Computational Fluid Dynamics (CFD)

- Numerically solves the Navier-Stokes and continuity equations
- Exact replication of structural form and surrounds.
- Generates explicit pressure or dynamic loads on structures that can be implemented in design calculations.....but do we trust them?
- Validation against wind tunnel and full-scale observation suggest mean results are OK, but peaks are poorly simulated.



Peng et al 2014



Dagnew and Gitsuamlak 2012

Designing for wind

Governing equations

Dynamic wind pressure

Dynamic response factor

$$p = (0.5\rho_{air})[V_{des,\theta}]^2 C_{shp} C_{dyn}$$

Pressure or unit force, calculated for sub-regions of a structure

Internal, external, friction

$$f = (0.5\rho_{air})[V_{des,\theta}]^2 C_{shp}$$
$$F = \sum ([P, f] \cdot A)$$

Force applied to action/member of interest

Tributary area for action/member of interest

Wind actions (AS/NZS1170.2)

Procedure

1. Determine site wind speeds
2. Determine design wind speed from the site wind speeds
3. Determine design wind pressures and/or forces (internal, external, dynamic)
4. Sum design pressures/forces to get required loading

Site wind speed ($V_{sit,\beta}$)

$$V_{sit,\beta} = V_R M_c M_d (M_{z,cat} M_s M_t)$$

$V_{sit,\beta}$ depends on:

Region (V_R [3.2])

- This must also consider average recurrence interval determined based on Importance Level

Climate change multiplier (M_c [3.4]) (statistical)

Direction (M_d [3.3]) (statistical)

Terrain roughness & height ($M_{z,cat}$ [4.2])

Shielding (M_s [4.3])

Topography (M_t [4.4]) (M_{lee} considered if in NZ)

Site wind speed ($V_{sit,\beta}$) must be calculated for 8 cardinal directions at reference height, h

Regional wind speed (V_R) [3.2]

- V_r describes the **gust wind speeds** with an **averaging period of 0.2 seconds** at **10 m elevation** over **flat, open terrain**.
- Australia and New Zealand are divided into 4 primary wind zones, A, B, C, D, with several sub-regions within these.
- Regions A and B1 are non-cyclonic, Regions B2, C and D are cyclonic.
- Return Period/ARI curves are presented for each region in tabular form.

Example:

Location - Brisbane

Importance Level 2

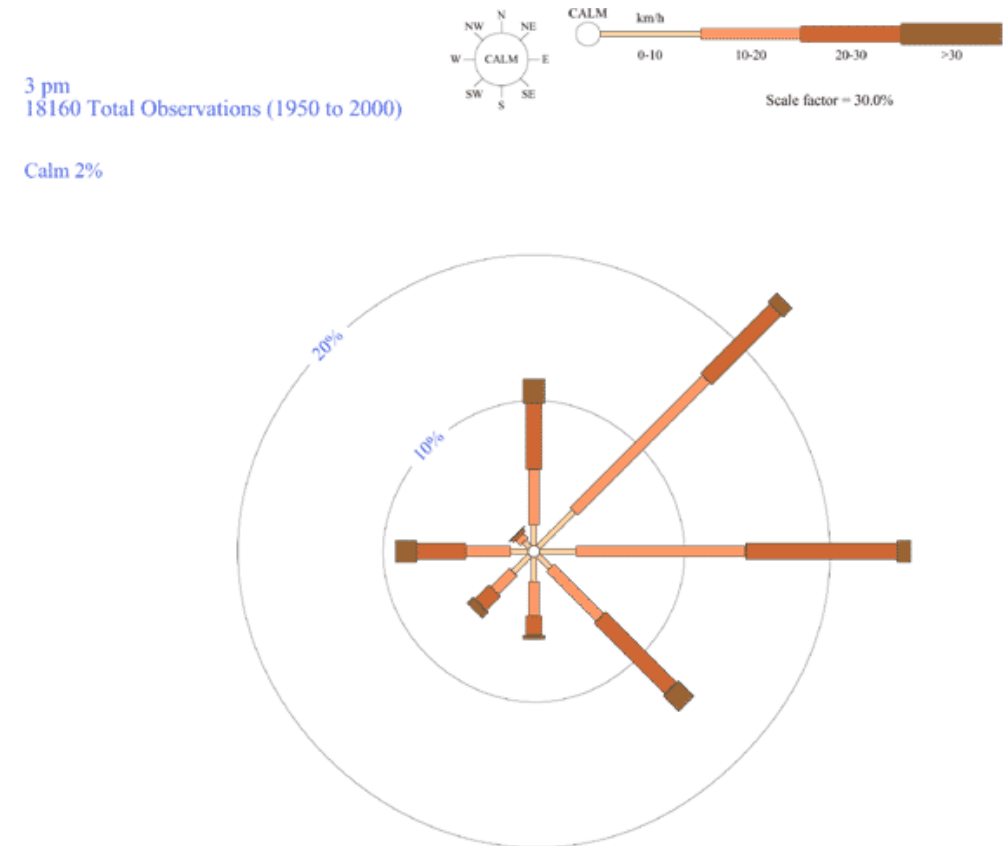
Design Life = 50 yr

ULS = 500 yr

SLS = 25 yr

Wind direction multiplier (M_d) [3.3]

- M_d is a statistical factor used to modify V_R in directions where strong winds are more or less likely.
- Taken as 1.0 for:
 - Chimneys, tanks, poles
 - Cladding and immediate supporting structures on buildings in Region B2, C, D
- Because M_d is a statistical factor it should not be used when standardising observations (e.g. in Assignment 2)

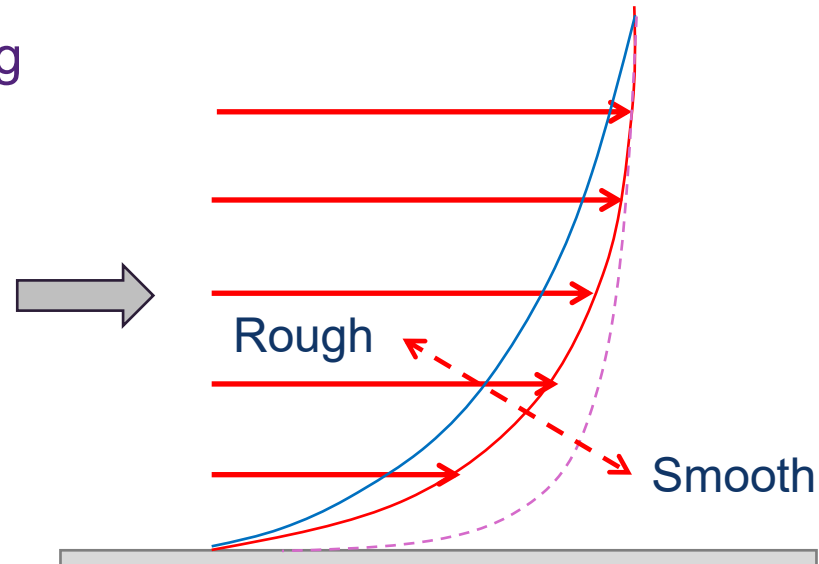


Climate Change multiplier (M_c) [3.4]

- Multiplier used to account for possible changes in extreme wind climate during the life of a structure being designed.
- Currently only cyclonic regions have an $M_c > 1.0$.

Terrain/height multiplier ($M_{z,cat}$) [4.2]

- Need to determine reference height (h)
- Consider terrain roughness upwind of building location (averaging generally required)
- $M_{z,cat}$ can be quite different for different directions



Terrain categories [4.2.1]



Terrain/height multiplier ($M_{z,cat}$) [4.2]

- Need to determine reference height (h)
- Consider terrain roughness upwind of building location (averaging generally required)
- $M_{z,cat}$ can be quite different for different directions

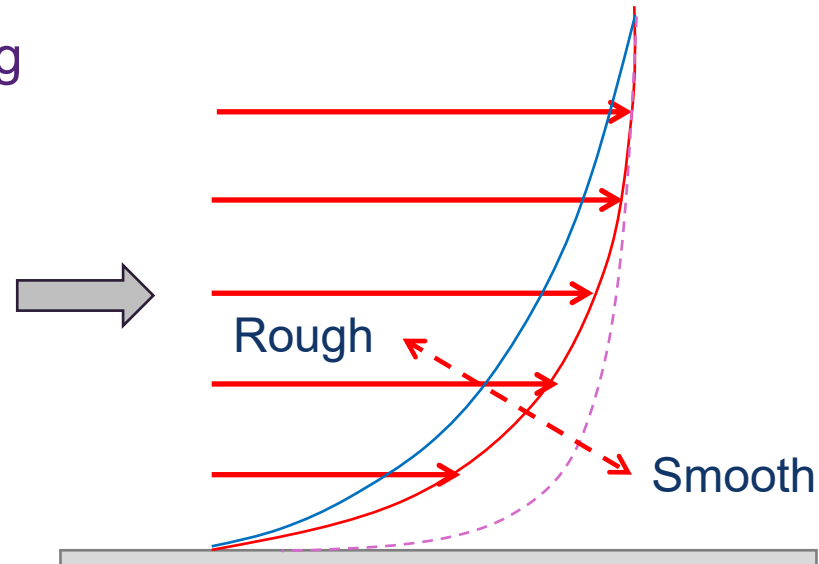
Example:

Location: Brisbane

Terrain Category 3

$h = 20$ m

$$M_{z,cat} = 0.94$$

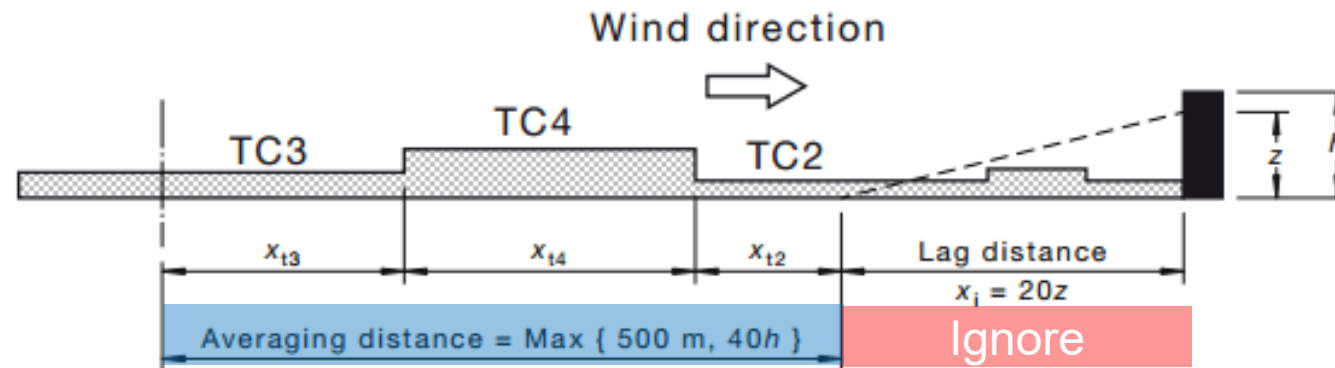


Averaging $M_{z,cat}$ [4.2.3]



Averaging $M_{z,cat}$ [4.2.3]

- Weighted average over distance
 $x_a = \max(500 \text{ m}, 40z)$
- Ignore upwind distance, $x_i = 20z$



$$M_{z,cat} = \frac{M_{z,cat2} x_{t2} + M_{z,cat4} x_{t4} + M_{z,cat3} x_{t3}}{x_{t2} + x_{t4} + x_{t3}} \text{ for the case illustrated}$$

Averaging $M_{z,cat}$ [4.2.3]

Example:

$z = 10 \text{ m}$

$h = 20 \text{ m}$

$M_{z,cat}$: E & W

Lag distance:

$X_i = 20z$

$= 20 \times 10 \text{ m}$

$= \underline{200 \text{ m}}$

Average distance:

$X_a = \max(500\text{m}, 40z)$

$= \underline{500 \text{ m}}$

Average $M_{z,cat}$:

$M_{z,cat-W} = (300 \times 1 + 200 \times 0.83) / 500$

$= \underline{0.93}$

$M_{z,cat-E} = 0.83$

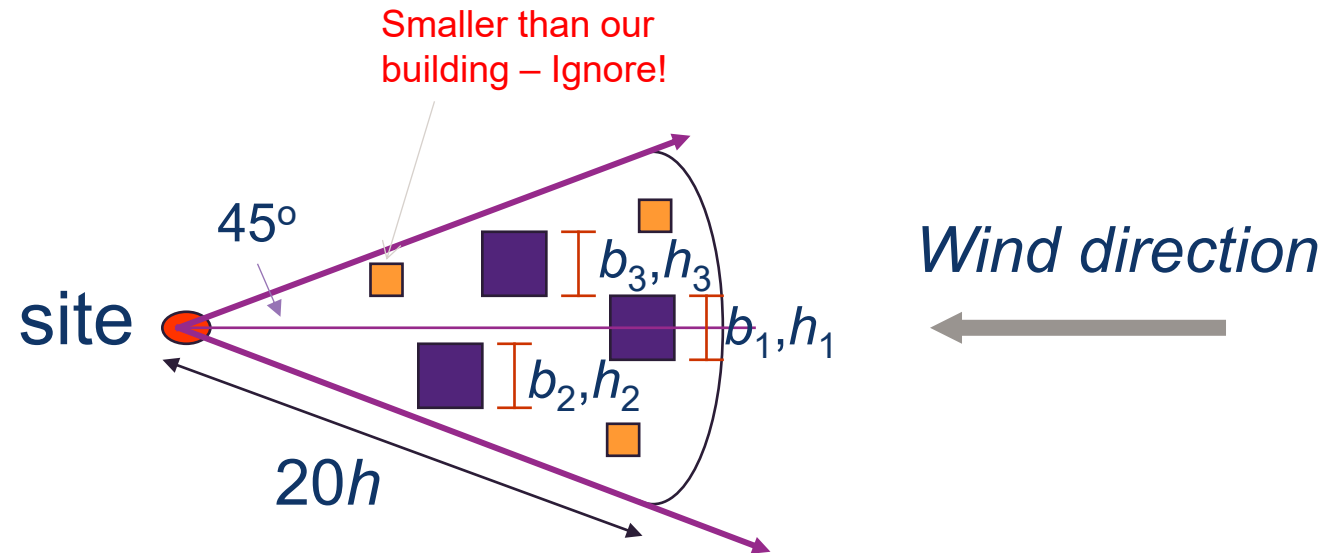


Shielding multiplier (M_s) [4.3]

- Upwind buildings can slow the wind speed seen by a given structure
- 1.0 where the average upwind ground gradient is > 0.2 and the overall height of the building does not exceed that of the design building
- 1.0 when building height is greater than 25 m – but wind tunnel testing is suggested
- Only buildings the same size or larger within $20h$ from the designed building can shield
- Vegetation/trees can not be considered

Shielding multiplier (M_s) [4.3]

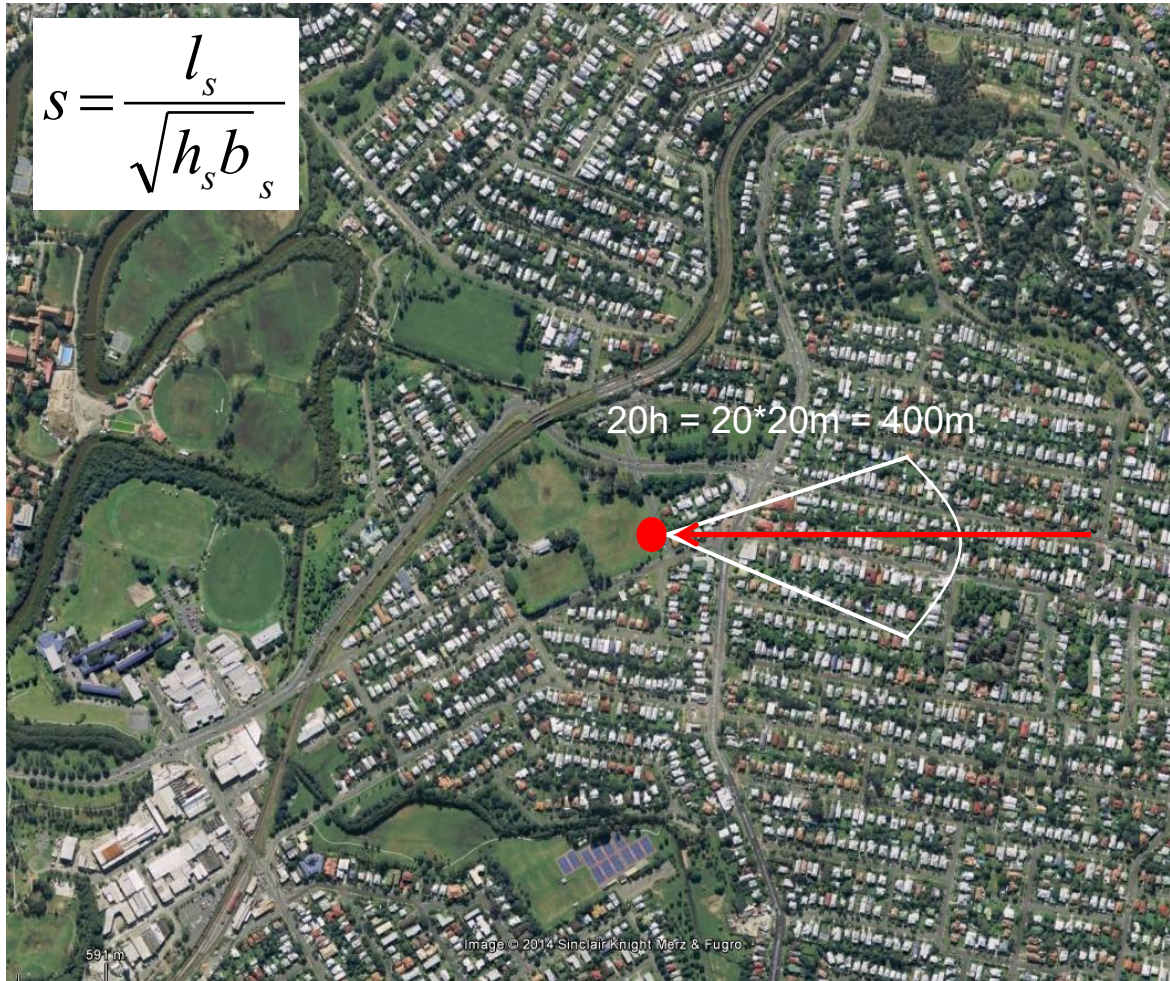
For each wind direction:



1. Evaluate average roof height
2. Evaluate average building width
3. Calculate shielding parameters, I_s and s .

Shielding multiplier (M_s) [4.3]

$$s = \frac{l_s}{\sqrt{h_s b_s}}$$



Example:

$h = 20$ m

M_s : E?

45° sector

Assume 60 bldgs. (70% $h = 6$ m,
10% $h = 12$ m, 20% $h = 25$ m).

All bldgs. $b = 10$ m.

$$\begin{aligned} l_s &= h(10/n_s + 5) \\ &= 20*(10/(0.2*60) + 5) \\ &= 116.7 \end{aligned}$$

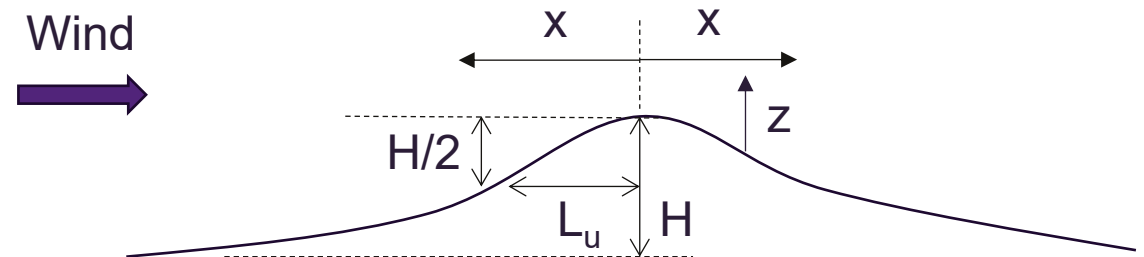
$$\begin{aligned} s &= 116.7/(25*10)^{0.5} \\ &= 7.4 \end{aligned}$$

$$M_s = 0.92$$

Topographic Multiplier (M_t) [4.4]

- Multiplier used to account for the influence of topography on wind speed.
- M_t depends on upwind slope and hill shape.
- Limited to 1.0 as a lower bound, so doesn't account for decrease in wind speed caused by topography (e.g. in valleys).
- Maximum value of 1.71.
- General equation:

$$M_h = 1 + \left(\frac{H}{3.5(z + L_1)} \right) \left(1 - \frac{|x|}{L_2} \right)$$



Topographic Multiplier (M_t) [4.4]

Australia

For Region A0 (thunderstorm dominated)

$$M_t = 0.5 + 0.5M_h$$

For other regions

$$M_t = M_h$$

Hill shape multiplier

New Zealand

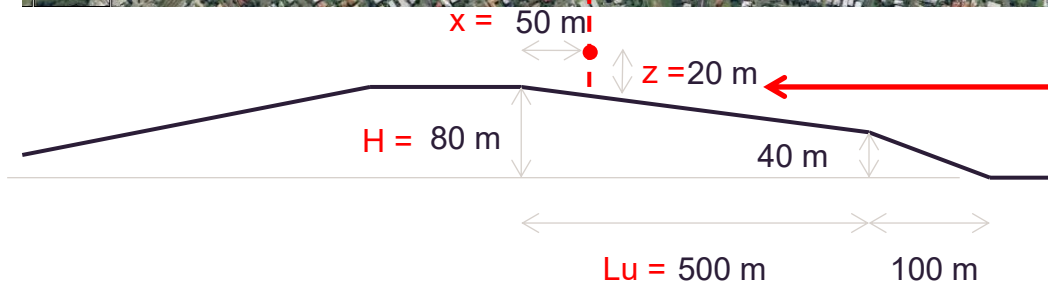
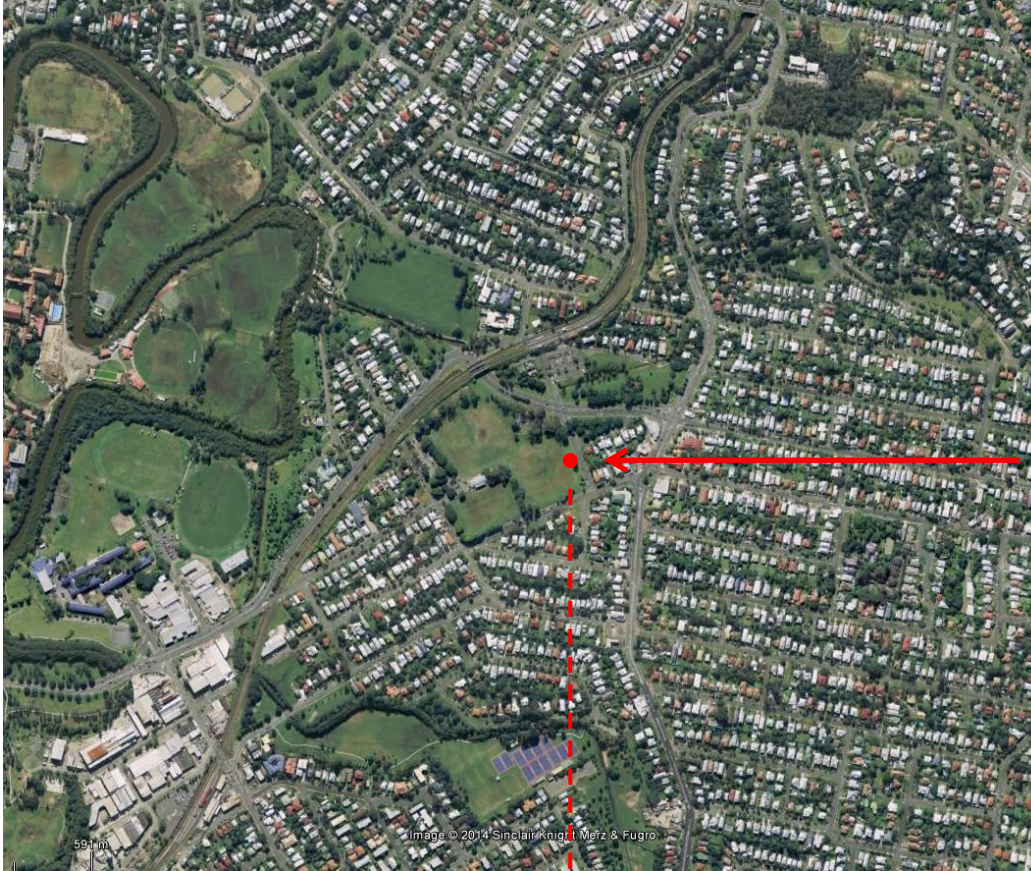
When elevation of building > 500 m above sea level

$$M_t = M_h M_{lee} (1 + 0.00015E)$$

Lee effect multiplier

Elevation ASL

Topographic Multiplier (M_t) [4.4]



Assume $z = 20$ m

$$\phi = H/2L_u = 80/(500 \cdot 2) = 0.08$$

$\phi > 0.05$, intermediate slope

$$M_h = 1 + \left(\frac{H}{3.5(z + L_1)} \right) \left(1 - \frac{|x|}{L_2} \right)$$

$$L_1 = \max(0.36L_u, 0.4H) = 180\text{m}$$

$$L_2 = 4L_1 = 720\text{m}$$

$$M_t = M_h = 1.11$$

Site wind speed



ULS, IL2, Brisbane, $h = 20\text{m}$,
Easterly wind.
Assume calc for major structural
element, $M_s = 0.92$
(*i.e. examples combined*)

$$V_{sit,\beta} = V_R M_c M_d (M_{z,cat} M_s M_t)$$

$$V_{sit,E} = 57 \times 1.0 \times 0.85 \times \dots$$

$(0.94 \times 0.92 \times 1.11)$

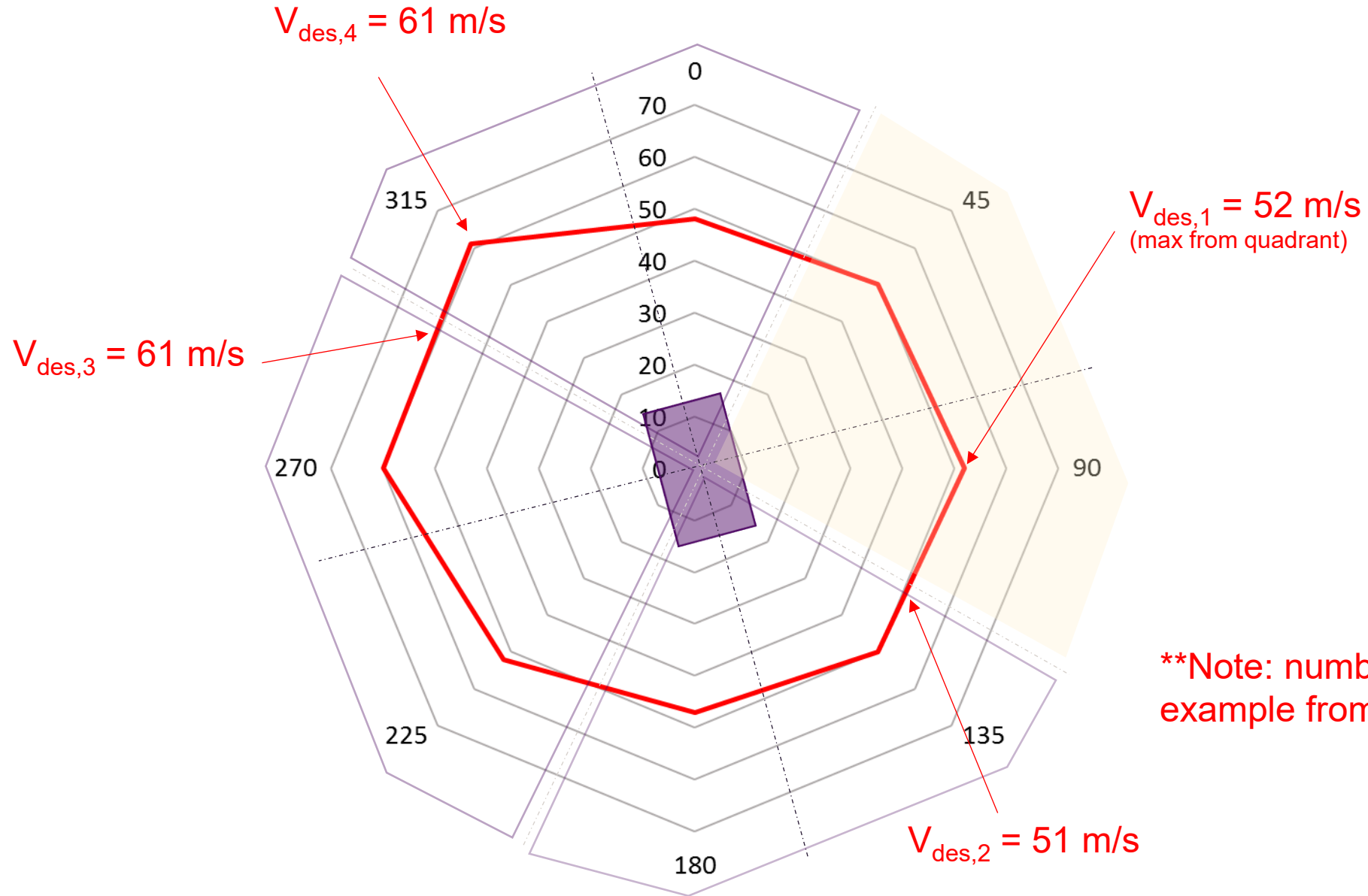
$$V_{sit,E} = 47 \text{ m/s}$$

Rounded to nearest m/s

Design wind speed ($V_{des,\theta}$)

- $V_{des,\theta}$ is equal to the maximum value of $V_{sit,\beta}$ in the range $\pm 45^\circ$ from a building's orthogonal axes.
- Therefore, depending on your structure, you may have 4 or more values of $V_{des,\theta}$.
- Conservatively it is acceptable to take the largest $V_{sit,\beta}$ value and use this for design from all directions.

Design wind speed ($V_{des,\theta}$)



AS4055 (2021)

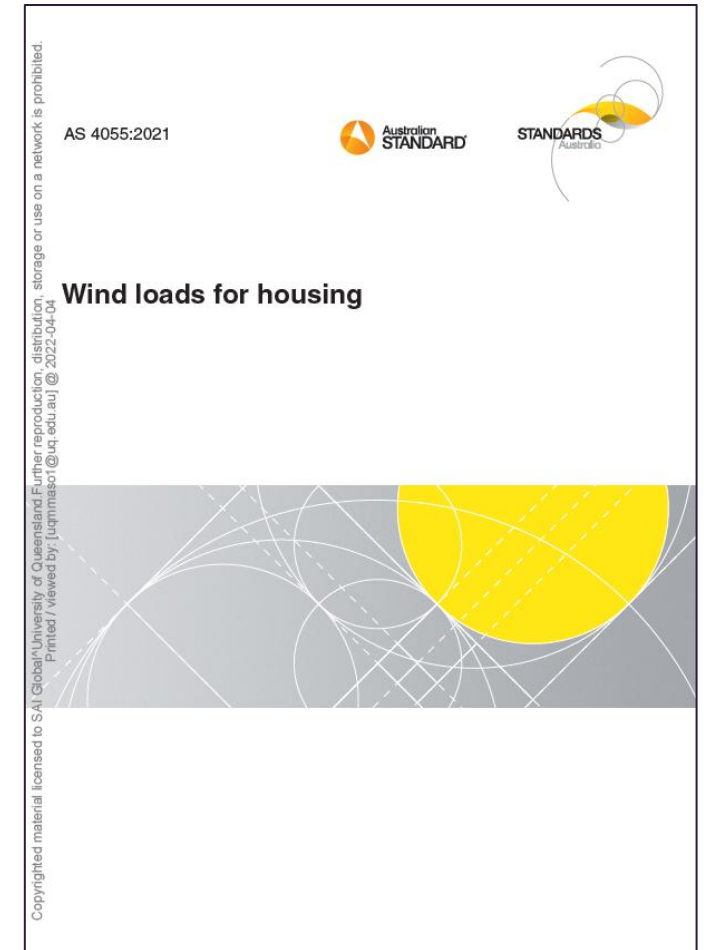
Specifies site wind speed classes for determining wind speeds and wind loads on NCC buildings Classes 1 & 10a (housing), within limits below.

Limits:

- Distance from ground to eaves < 6 m
- Distance from ground to top of roof < 8.5 m
- Building width (W) must be < 16 m (max eaves 0.9 m)
- Building length (L) to width (W) ratio must be < 5
- Roof pitch $< 35^\circ$

Outside limits – use AS/NZS1170.2

AS4055 and AS/NZS1170.2 do not match, so you can only use one for each design (the latter is more complete).



AS4055 (2021)

Key differences to AS/NZS1170.2 (*wind speeds only*)

- Uses a simplified tabular design approach,
- Use of wind classes (N or C – non-cyclonic, cyclonic),
- No explicit reference to height,
- No explicit reference to building importance or wind speed return periods (all assumed).

Similarities (*wind speeds only*)

- Same wind regions (i.e. A, B, C, D),
- Same terrain categories (i.e. 1-3),
- Considers influence of topography, terrain, shielding (no directional multiplier though).

AS4055 (2021)

- General process [2.101]
- Wind region [2.2]
- Terrain classification [2.3]
- Topographic classification [2.4]
- Shielding class [2.5]
- Wind class [Table 2.2]

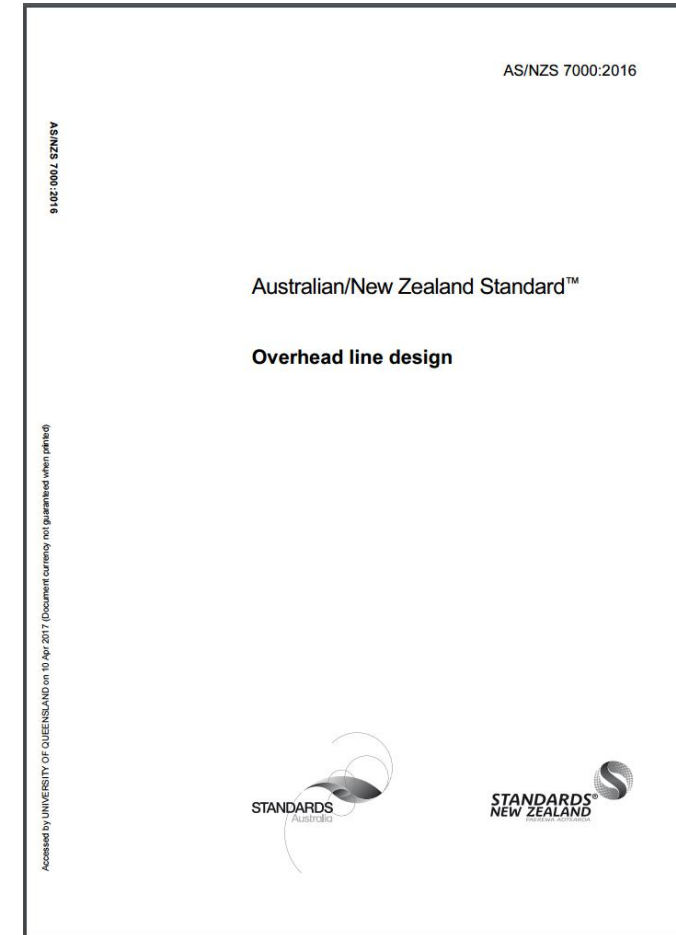
AS/NZS7000 (2016)

Overarching standard for design of overhead power lines.

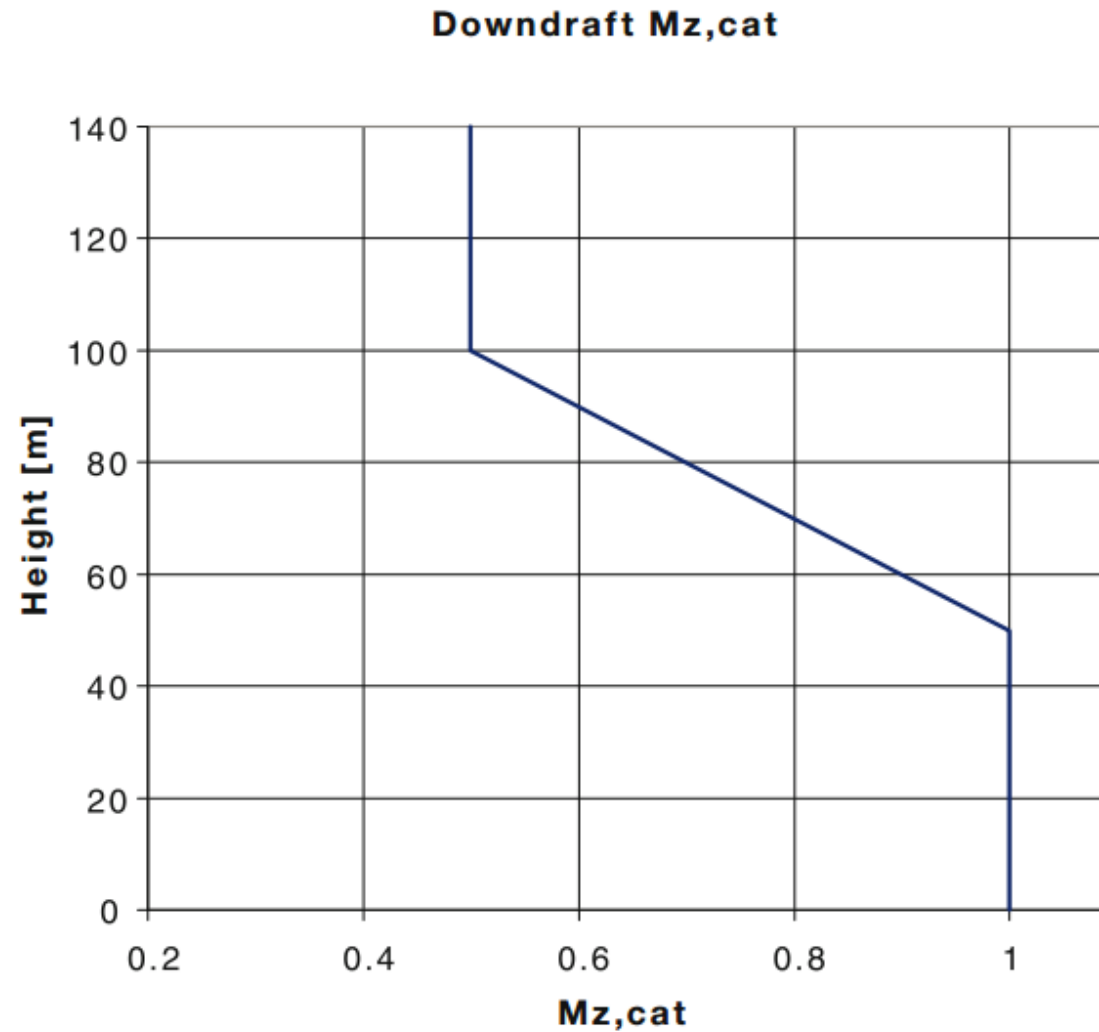
Appendix B of interest for wind load calculations.

This standard is of interest because it explicitly considers thunderstorm generated winds.

- Wind zones specified to determine if thunderstorm calculations are needed.
- Only uses Region A magnitudes.
- No thunderstorm wind calcs for Regions C & D.
- Thunderstorm $M_{Z,cat}$, M_t and SRF provided.



Terrain height multiplier ($M_{z,cat}$)



Topographic multiplier (M_t)

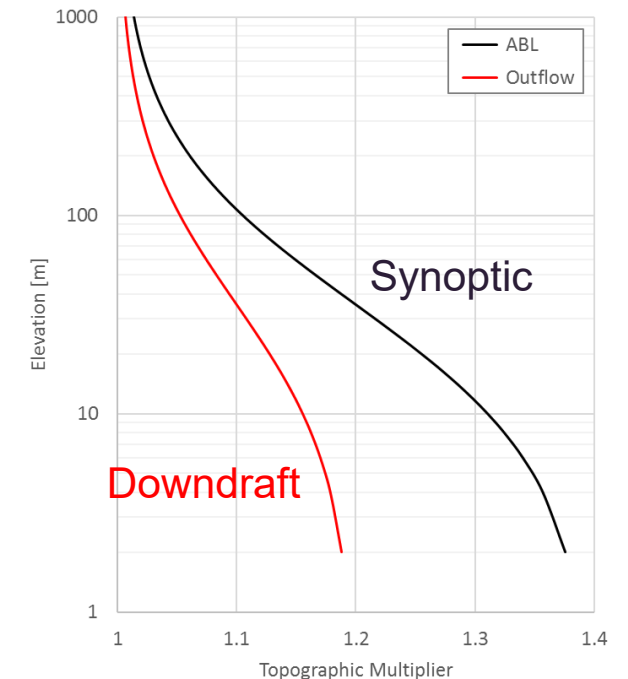
Research has shown that thunderstorm generated winds may be influenced to a lesser extent than synoptic winds.

Thunderstorm generated winds topographic multiplier, $M_{t,downdraft}$, provided.

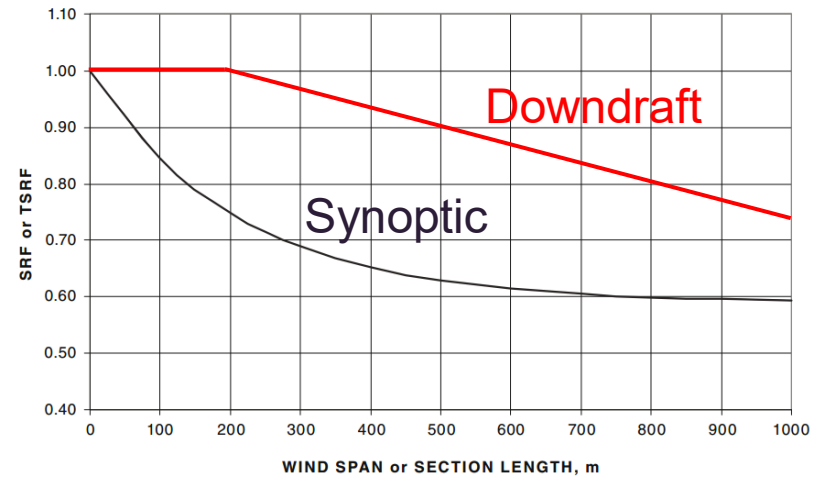
$$M_{t,downdraft} = 0.5 + 0.5M_{t,synoptic}$$

M_h from AS/NZS1170.2

****Region A0 in AS/NZS1170.2
now matches this approach****



Span reduction factor

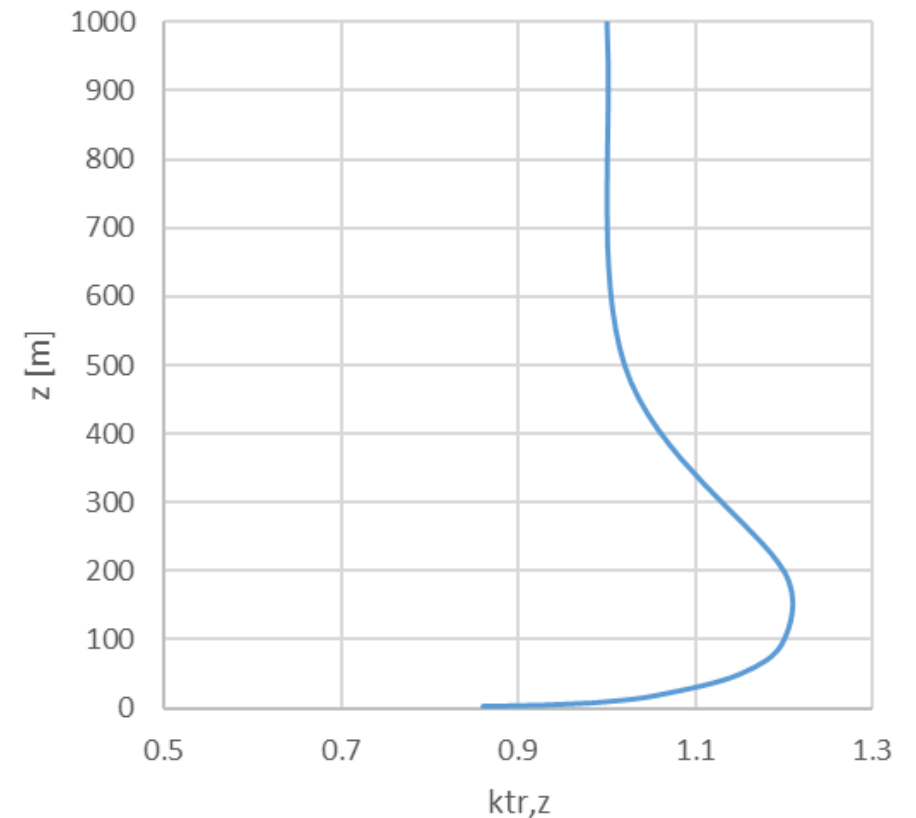


Span Reduction Factors

This is a measure of correlation in the wind

ISO4354 (2009)

- Two key contributions (beyond AS/NZS1170.2)
- Provides a method for conversion between averaging times (gust and mean).
- Provides a thunderstorm generated wind gust wind speed profile.



Summary

- Reviewed several major wind loading standards and how they deal with wind speed calculations.
- Been through some examples of the application of AS/NZS1170.2 and AS4055.
- Seen how more progressive standards have begun to incorporate thunderstorm generated winds.

Contact

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